



EEE 1101

Electrical Circuits

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Course Content

Unbalanced Polyphase Circuits:

Unbalanced wye and delta loads, Network solutions, Phase-sequence effects, Methods of checking voltage phase sequence, Power measurements (both active and reactive) in unbalanced three phase systems.

Reference Book

Alternating Current Circuits (4th edition)

- Russel M.kerchner
- George F.Corcoran

Unbalanced Polyphase Circuits



Any polyphase load in which the impedance in one or more phases differs from those of other phases is said to be unbalanced. Even though the load impedances of the various phases are identical, one of the methods of calculating unbalanced loads must be employed if the voltages impressed on the load are unequal and differ in phase by angles which are not equal.

Unbalanced Polyphase Circuits



Delta Connected Unbalanced Load :

Example 1. Given the unbalanced delta load shown in Fig. 1. Calculate all currents for the three-phase balanced voltages shown on the figure, if the voltage sequence is ab–ca–bc.

Solution:

$$V_{ab} = 100\angle 0^\circ \text{V}, V_{bc} = 100\angle 120^\circ \text{V}, V_{ca} = 100\angle -120^\circ \text{V}$$

Phase Currents:

$$I_{ab} = \frac{100\angle 0^\circ}{6+j8} = 10\angle -53.1^\circ \text{A}$$

$$I_{bc} = \frac{100\angle 120^\circ}{4-j3} = 20\angle 156.9^\circ \text{A}$$

$$I_{ca} = \frac{100\angle -120^\circ}{20} = 5\angle -120^\circ \text{A}$$

Line Currents:

$$I_{a'a} = I_{ab} - I_{ca} = 9.26\angle -23.32^\circ \text{A}$$

$$I_{b'b} = I_{bc} - I_{ab} = 29\angle 147^\circ \text{A}$$

$$I_{c'c} = I_{ca} - I_{bc} = 20\angle -37.4^\circ \text{A}$$

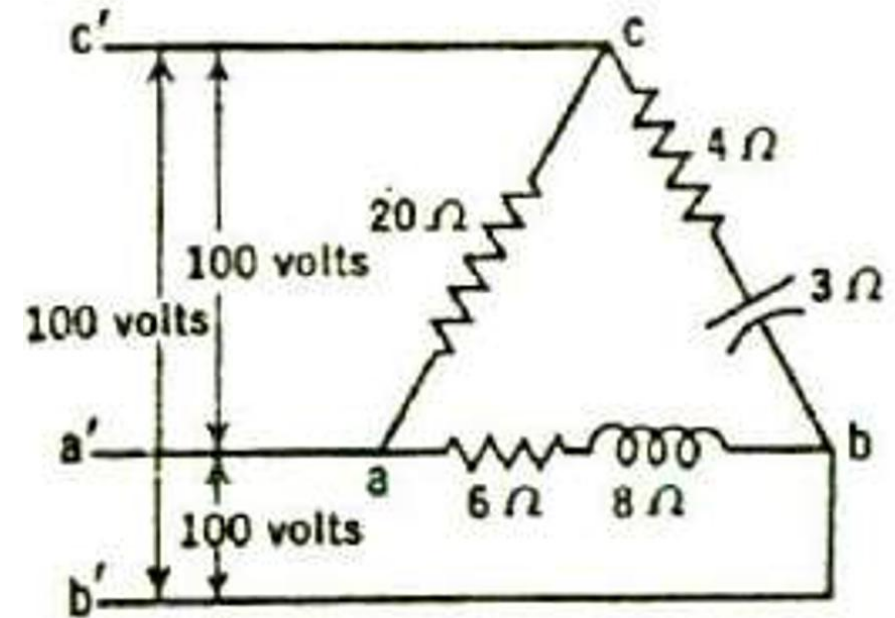


FIG. 1. Unbalanced delta load.
See example 1.

Unbalanced Polyphase Circuits



Three Wire Star Connected Unbalanced Load :

Example 2. A balanced set of three-phase voltages is connected to an unbalanced set of wye-connected impedances as shown in Fig. 2. The following values are assumed to be known:

$$V_{ab} = 212 \angle 90^\circ \text{ volts}$$

$$V_{bc} = 212 \angle -150^\circ \text{ volts}$$

$$V_{ca} = 212 \angle -30^\circ \text{ volts}$$

$$Z_{an} = 10 + j0 \text{ ohms}$$

$$Z_{bn} = 10 + j10 \text{ ohms}$$

$$Z_{cn} = 0 - j20 \text{ ohms}$$

The line currents $I_{a'a}$, $I_{b'b}$, and $I_{c'c}$ are to be determined by the wye to delta conversion method.

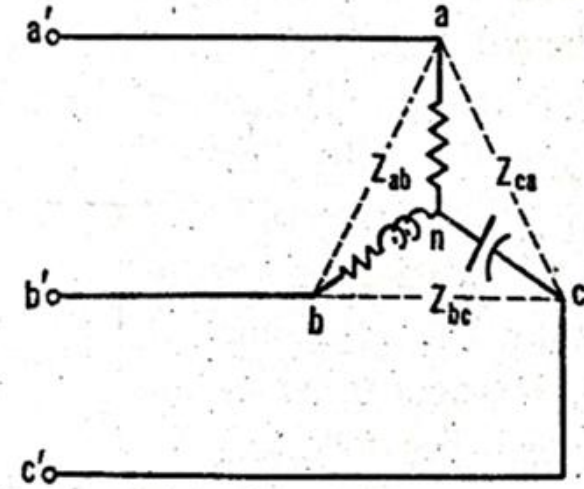


FIG. 2. Conversion from a wye-connected load to an equivalent delta-connected load.

Solution:

Numerically, the equivalent delta impedances are:

$$Z_{ab} = \frac{300 - j300}{0 - j20} = (15 + j15) = 21.2 \angle 45^\circ \text{ ohms}$$

$$Z_{bc} = \frac{300 - j300}{10 - j0} = (30 - j30) = 42.4 \angle -45^\circ \text{ ohms}$$

$$Z_{ca} = \frac{300 - j300}{10 + j10} = (0 - j30) = 30.0 \angle -90^\circ \text{ ohms}$$

$$Z_{ab} = \frac{(Z_{an}Z_{bn} + Z_{bn}Z_{cn} + Z_{cn}Z_{an})}{Z_{nc}} = \frac{S}{Z_{nb}}$$

$$Z_{bc} = \frac{S}{Z_{cn}} \quad \text{and} \quad Z_{ca} = \frac{S}{Z_{bn}}$$

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The load currents in the equivalent delta are:

$$I_{ab} = \frac{V_{ab}}{Z_{ab}} = \frac{212 \angle 90^\circ}{21.2 \angle 45^\circ} = 10 \angle 45^\circ \text{ amperes}$$

$$I_{bc} = \frac{V_{bc}}{Z_{bc}} = \frac{212 \angle -150^\circ}{42.4 \angle -45^\circ} = 5.0 \angle -105^\circ \text{ amperes}$$

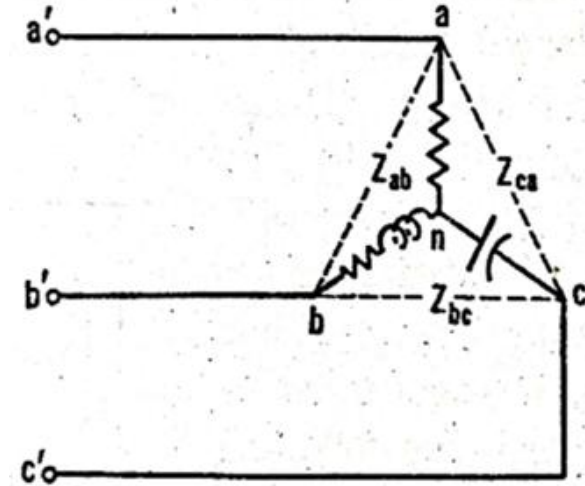
$$I_{ca} = \frac{V_{ca}}{Z_{ca}} = \frac{212 \angle -30^\circ}{30 \angle -90^\circ} = 7.07 \angle 60^\circ \text{ amperes}$$

The actual line and load currents are:

$$\begin{aligned} I_{a'a} &= I_{ab} - I_{ca} \\ &= 10 \angle 45^\circ - 7.07 \angle 60^\circ = 3.66 \angle 15^\circ \text{ amperes} \end{aligned}$$

$$\begin{aligned} I_{b'b} &= I_{bc} - I_{ab} \\ &= 5 \angle -105^\circ - 10 \angle 45^\circ = 14.56 \angle -125.1^\circ \text{ amperes} \end{aligned}$$

$$\begin{aligned} I_{c'c} &= I_{ca} - I_{bc} \\ &= 7.07 \angle 60^\circ - 5 \angle -105^\circ = 11.98 \angle 68.2^\circ \text{ amperes} \end{aligned}$$



Unbalanced Polyphase Circuits



Example 2. A balanced set of three-phase voltages is connected to an unbalanced set of wye-connected impedances as shown in Fig. 2. The following values are assumed to be known:

$$V_{ab} = 212 \angle 90^\circ \text{ volts}$$

$$V_{bc} = 212 \angle -150^\circ \text{ volts}$$

$$V_{ca} = 212 \angle -30^\circ \text{ volts}$$

$$Z_{an} = 10 + j0 \text{ ohms}$$

$$Z_{bn} = 10 + j10 \text{ ohms}$$

$$Z_{cn} = 0 - j20 \text{ ohms}$$

The line currents $I_{a'a}$, $I_{b'b}$, and $I_{c'c}$ are to be determined by the wye to delta conversion method.

Solution:

Using mesh analysis,

$$(Z_{an} + Z_{bn})I_1 - Z_{bn}I_2 = V_{ab}$$

$$\Rightarrow (10 + 10 + j10)I_1 - (10 + j10)I_2 = 212 \angle 90^\circ$$

$$\Rightarrow (20 + j10)I_1 - (10 + j10)I_2 = 212 \angle 90^\circ \dots\dots\dots (i)$$

$$(Z_{bn} + Z_{cn})I_2 - Z_{bn}I_1 = V_{bc}$$

$$\Rightarrow (10 + j10 - j20)I_2 - (10 + j10)I_1 = 212 \angle -150^\circ$$

$$\Rightarrow -(10 + j10)I_1 + (10 - j10)I_2 = 212 \angle -150^\circ \dots\dots\dots (ii)$$

Solving equation (i) & (ii),

$$I_1 = 3.65 \angle 15^\circ \text{ A and } I_2 = 11.96 \angle -113.8^\circ \text{ A}$$

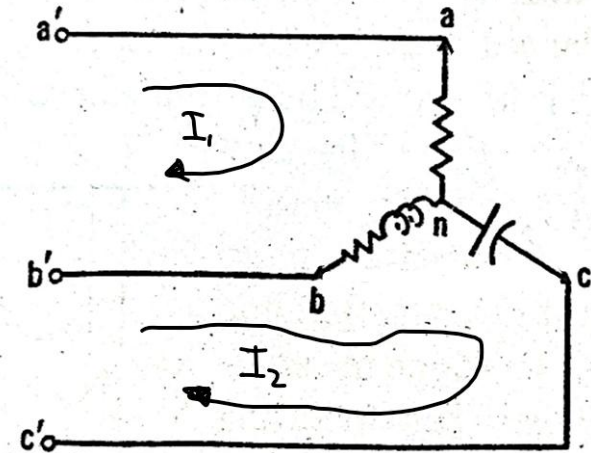


FIG. 2. Conversion from a wye-connected load to an equivalent delta-connected load.

Line currents:

$$I_{a'a} = I_1 = 3.65 \angle 15^\circ \text{ A}$$

$$I_{c'c} = -I_2 = -11.96 \angle -113.8^\circ \text{ A} = 11.96 \angle 66.2^\circ \text{ A}$$

$$I_{b'b} = -(I_{a'a} + I_{c'c}) = 14.52 \angle -125.1^\circ \text{ A}$$

Unbalanced Polyphase Circuits



Four Wire Star Connected Unbalanced Load :

Q. Non-reactive loads of 10 kW, 8 kW and 5 kW are connected between the neutral and the red, yellow and blue phases respectively of a 3-phase, 4-wire system. The line voltage is 400V. Calculate (i) the current in each line and (ii) the current in the neutral wire.

Solution:

Magnitude of Phase voltage, $V_p = V_L/\sqrt{3} = 400/\sqrt{3} = 231 \text{ V}$

Magnitude of Phase and line Currents:

$$I_R = 10000/231 = 43.3 \text{ A}$$

$$I_Y = 8000/231 = 34.6 \text{ A}$$

$$I_B = 5000/231 = 21.65 \text{ A}$$

Considering Positive phase sequence,

Phase Voltages and Currents in Complex Vector Form:

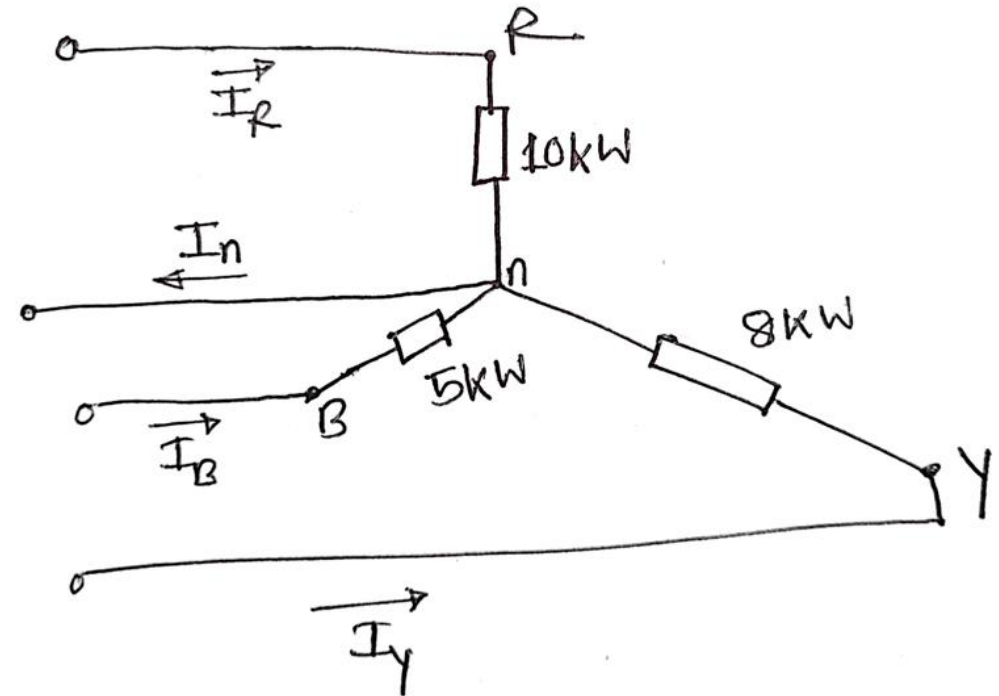
$$V_{Rn} = 231\angle 0^\circ \text{ V}, V_{Yn} = 231\angle -120^\circ \text{ V}, V_{Bn} = 231\angle 120^\circ \text{ V}$$

$$I_R = 43.3\angle 0^\circ \text{ A}, I_Y = 34.6\angle -120^\circ \text{ A}, I_B = 21.65\angle 120^\circ \text{ A}$$

Neutral Current,

$$I_n = I_R + I_Y + I_B = 43.3\angle 0^\circ + 34.6\angle -120^\circ + 21.65\angle 120^\circ$$

$$\Rightarrow I_n = 18.86\angle -36.46^\circ \text{ A}$$



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Example 3: The generated voltages and impedances for Fig. 5 are given as follows:

$$E_{an} = 1000\angle 0^\circ, E_{bn} = 1000\angle -120^\circ, E_{cn} = 1000\angle -240^\circ,$$

$$Z_{na} = 2+j8, Z_{aa'} = 1+j2, Z_{a'n'} = 19+j18,$$

$$Z_{nb} = 2+j8, Z_{bb'} = 1+j2, Z_{b'n'} = 49-j2,$$

$$Z_{nc} = 2+j8, Z_{aa'} = 1+j2 \text{ and } Z_{c'n'} = 29+j50.$$

Find the line currents, phase voltages, and line voltages at the load end.

Solution:

$$Z_a = 2+j8+1+j2+19+j18 = 22+j28 \Omega$$

$$Z_b = 2+j8+1+j2+49-j2 = 52+j8 \Omega$$

$$Z_c = 2+j8+1+j2+29+j50 = 32+j60 \Omega$$

Using mesh analysis,

$$(Z_a + Z_b)I_1 - Z_b I_2 = E_{an} - E_{bn}$$

$$\Rightarrow (22+j28 + 52+j8) I_1 - (52+j8) I_2 = 1000\angle 0^\circ - 1000\angle -120^\circ$$

$$\Rightarrow (74+j36) I_1 - (52+j8) I_2 = 1500+j866 \dots\dots\dots(i)$$

$$(Z_b + Z_c)I_2 - Z_b I_1 = E_{bn} - E_{cn}$$

$$\Rightarrow (52+j8 + 32+j60) I_2 - (52+j8) I_1 = 1000\angle -120^\circ - 1000\angle -240^\circ$$

$$\Rightarrow - (52+j8) I_1 + (84+j68) I_2 = -j1732 \dots\dots\dots(ii)$$

Solving equation (i) & (ii),

$$I_1 = 16\angle -34.9^\circ \text{ A and } I_2 = 20.7\angle -109.2^\circ \text{ A}$$

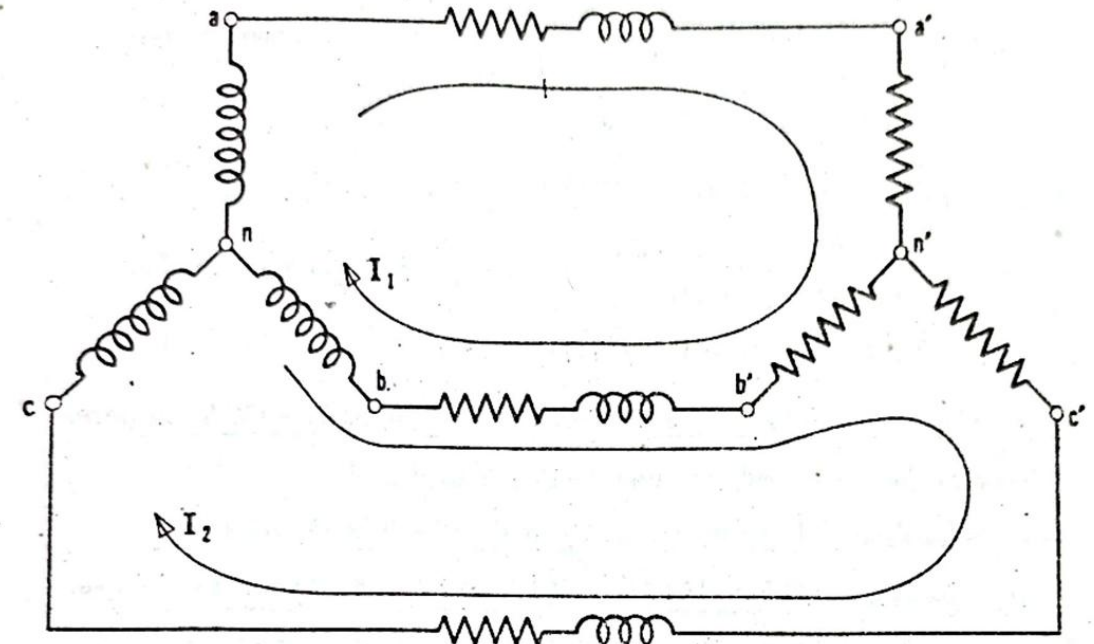


FIG. 5.

Unbalanced Polyphase Circuits



Line currents:

$$I_{aa'} = I_1 = 16 \angle -34.9^\circ \text{ A}$$

$$I_{cc'} = -I_2 = -20.7 \angle -109.2^\circ \text{ A} = 20.7 \angle 70.8^\circ \text{ A}$$

$$I_{bb'} = -(I_{aa'} + I_{cc'}) = 22.47 \angle -152.4^\circ \text{ A}$$

Phase voltages at load end :

$$V_{a'n'} = 16 \angle -34.9^\circ \times (19 + j18) = 418.76 \angle 8.5^\circ \text{ V}$$

$$V_{b'n'} = 22.47 \angle -152.4^\circ \times (49 - j2) = 1102.32 \angle -154.8^\circ \text{ V}$$

$$V_{c'n'} = 20.7 \angle 70.8^\circ \times (29 + j50) = 1196.48 \angle 130.7^\circ \text{ V}$$

Line voltages at load end :

$$V_{a'b'} = V_{a'n'} - V_{b'n'} = 418.76 \angle 8.5^\circ - 1102.32 \angle -154.8^\circ = 1508.2 \angle 20.6^\circ \text{ V}$$

$$V_{b'c'} = V_{b'n'} - V_{c'n'} = 1102.32 \angle -154.8^\circ - 1196.48 \angle 130.7^\circ = 1393.46 \angle -98.9^\circ \text{ V}$$

$$V_{c'a'} = V_{c'n'} - V_{a'n'} = 1196.48 \angle 130.7^\circ - 418.76 \angle 8.5^\circ = 1463.1 \angle 144^\circ \text{ V}$$

Assignment: A 3-phase, 4-wire system supplies power at 400 V and lighting at 230 V. If the lamps in use require 65, 80 and 30 amperes in each of the three lines, what should be the current in the neutral wire? If a 3-phase motor is now started, taking 220 A from the lines at a p.f. of 0.25 lagging, what should be the total current in each line and the neutral wire? Find also the total power supplied to the lamps and the motor.

Unbalanced Polyphase Circuits



The Y-Y System with Neutral Connection:

Line currents:

$$I_{a'a} = \frac{E_{n'a'} - I_{nn'}Z_n}{(Z_g + Z_l + Z_{an})}; \quad I_{b'b} = \frac{E_{n'b'} - I_{nn'}Z_n}{(Z_g + Z_l + Z_{bn})};$$

$$I_{c'c} = \frac{E_{n'c'} - I_{nn'}Z_n}{(Z_g + Z_l + Z_{cn})}$$

Let,

$$Z_g + Z_l + Z_{an} = Z_a$$

$$Z_g + Z_l + Z_{bn} = Z_b$$

$$Z_g + Z_l + Z_{cn} = Z_c$$

Applying KCL at neutral point,

$$I_{a'a} + I_{b'b} + I_{c'c} = I_{nn'}$$

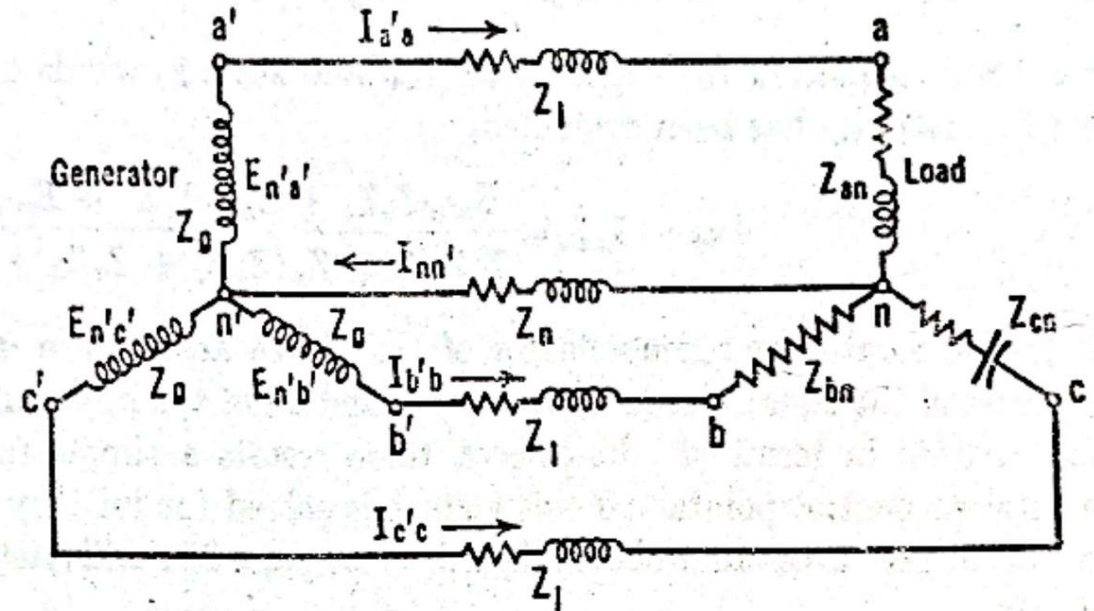


FIG. 8. A four-wire three-phase system.

Unbalanced Polyphase Circuits



$$I_{a'a} + I_{b'b} + I_{c'c} = I_{nn'}$$

$$\Rightarrow \frac{E_{n'a'} - I_{nn'}Z_n}{Z_a} + \frac{E_{n'b'} - I_{nn'}Z_n}{Z_b} + \frac{E_{n'c'} - I_{nn'}Z_n}{Z_c} = I_{nn'}$$

$$\Rightarrow \frac{E_{n'a'}Z_bZ_c - I_{nn'}Z_nZ_bZ_c + E_{n'b'}Z_aZ_c - I_{nn'}Z_aZ_c + E_{n'c'}Z_aZ_b - I_{nn'}Z_nZ_aZ_b}{Z_aZ_bZ_c} = I_{nn'}$$

$$\Rightarrow E_{n'a'}Z_bZ_c + E_{n'b'}Z_aZ_c + E_{n'c'}Z_aZ_b = I_{nn'}[Z_aZ_bZ_c + Z_n(Z_bZ_c + Z_cZ_a + Z_aZ_b)]$$

$$I_{nn'} = \frac{E_{n'a'}Z_bZ_c + E_{n'b'}Z_cZ_a + E_{n'c'}Z_aZ_b}{Z_aZ_bZ_c + Z_n(Z_bZ_c + Z_cZ_a + Z_aZ_b)}$$

Balanced Polyphase Circuits



Determination of Phase sequence:

1. Two lamp method:

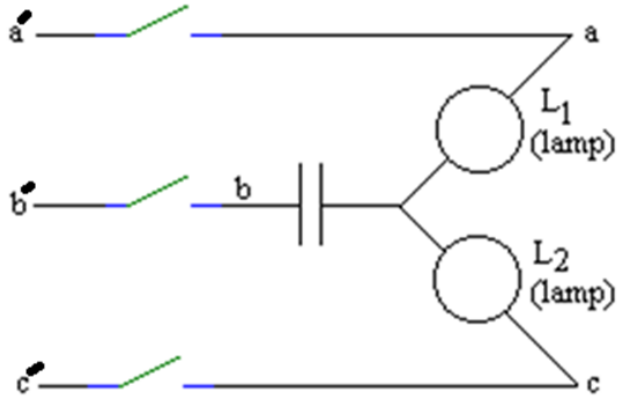
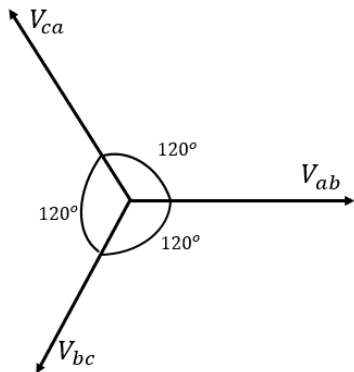
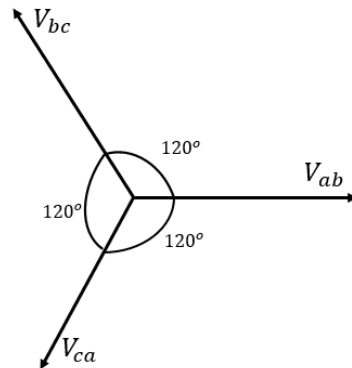


Fig. 1: Circuit for two lamp method

- The lamps L_1 and L_2 should be similar.
- If lamp L_1 is brighter than L_2 the phase sequence is ab-ca-bc.
- If lamp L_2 is brighter than L_1 , the phase sequence is ab-bc-ca.



ab-bc-ca or a-b-c sequence



ab-ca-bc or a-c-b sequence

2. Voltmeter method:

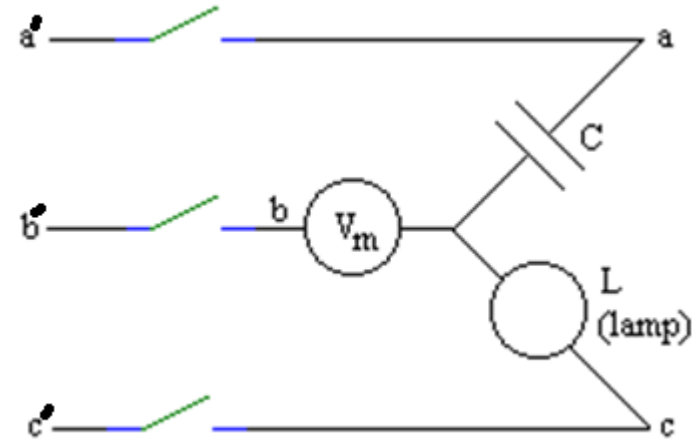


Fig. 2: Circuit for voltmeter method

- Capacitive reactance should be greater than or equal to lamp-board resistance.
- If voltmeter V_m reads above the line voltage then voltage sequence is ab-bc-ca
- If voltmeter reads below the line voltage then the phase sequence is ab-ca-bc.



Unbalanced Polyphase Circuits

Methods of Checking voltage phase sequence :

Q. A 220 V, three-phase supply with phase sequence ab–bc–ca is applied to the circuit shown in the fig. 1. The lamps are rated at 230 V, 100 W each, and the capacitor is 10 μ F. Prove that Lamp(L_2) is brighter than Lamp(L_1) .

Solution:

$$V_{ab} = 220\angle 0^\circ \text{V} , V_{bc} = 220\angle -120^\circ \text{V} , V_{ca} = 220\angle -240^\circ \text{V}$$

Using mesh analysis,

$$(Z_{an} + Z_{bn})I_1 - Z_{bn}I_2 = V_{ab}$$

$$\Rightarrow (529 - j318.3)I_1 - (-j318.3)I_2 = 220\angle 0^\circ$$

$$\Rightarrow (529 - j318.3)I_1 + (j318.3)I_2 = 220\angle 0^\circ \dots\dots\dots(i)$$

$$(Z_{bn} + Z_{cn})I_2 - Z_{bn}I_1 = V_{bc}$$

$$\Rightarrow (529 - j318.3)I_2 - (-j318.3)I_1 = 220\angle -120^\circ$$

$$\Rightarrow (j318.3)I_1 + (529 - j318.3)I_2 = 220\angle -120^\circ \dots\dots\dots(ii)$$

Solving equation (i) & (ii),

$$I_1 = 0.15\angle 18.1^\circ \text{A} \text{ and } I_2 = 0.41\angle -80.9^\circ \text{A}$$

Line currents:

$$I_{aa'} = I_1 = 0.15\angle 18.1^\circ \text{A}$$

$$I_{cc'} = -I_2 = -0.41\angle -80.9^\circ \text{A} = 0.41\angle 99.1^\circ \text{A}$$

$$I_{bb'} = -(I_{aa'} + I_{cc'}) = 0.45\angle -99.76^\circ \text{A}$$

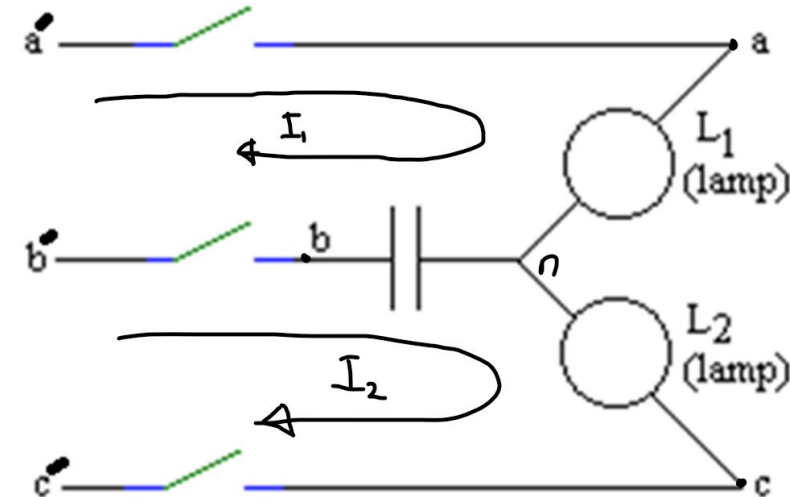


fig.1

$$Z_{an} = Z_{cn} = 230^2/100 = 529\Omega$$

$$Z_{nb} = -j/(2\pi \times 50 \times 10 \times 10^{-6})\Omega = -j318.3\Omega$$

Phase voltages at load end :

$$V_{an} = 0.15\angle 18.1^\circ \times 529 = 79.35\angle 18.1^\circ \text{V}$$

$$V_{bn} = 0.45\angle -99.76^\circ \times (-j318.3) = 143.23\angle 170.2^\circ \text{V}$$

$$V_{cn} = 0.41\angle 99.1^\circ \times 529 = 216.9\angle 99.1^\circ \text{V}$$

Power dissipation of Lamps:

$$P_1 = 79.35^2/529 = 11.9 \text{ W}$$

$$P_2 = 216.9^2/529 = 88.93 \text{ W}$$

Unbalanced Polyphase Circuits



Methods of Checking voltage phase sequence :

Q. A 220 V, three-phase supply with phase sequence ab–ca–bc is applied to the circuit shown in the fig. 1. The lamps are rated at 230 V, 100 W each, and the capacitor is $10\ \mu\text{F}$. Prove that Lamp(L_1) is brighter than Lamp(L_2) .

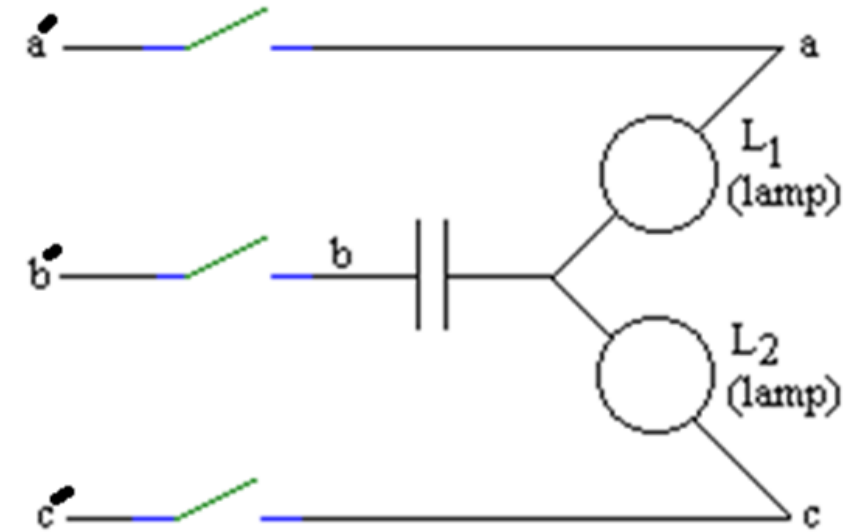


fig.1

Unbalanced Polyphase Circuits



Methods of Checking voltage phase sequence :

Q. A 200 V three-phase supply with phase sequence ab–bc–ca is applied to the circuit shown in Fig. 1. The lamp is rated at 230 V, 100 W, and the capacitor has a capacitance of 10 μ F. Show that the voltmeter reading is greater than the line voltage.

Solution:

$$V_{ab} = 200\angle 0^\circ \text{ V}, V_{bc} = 200\angle -120^\circ \text{ V}, V_{ca} = 200\angle 120^\circ \text{ V}$$

$$Z_{an} = -j/(2\pi \times 50 \times 10 \times 10^{-6}) \Omega = -j318.3 \Omega$$

$$Z_{cn} = 230^2/100 = 529 \Omega$$

$$V_{ac} = 200\angle -60^\circ \text{ V}$$

$$I_1 = (200\angle -60^\circ)/(529 - j318.3) = 0.324\angle -29^\circ \text{ A}$$

$$V_{ab} = V_{an} - V_{bn} = I_1 Z_{cn} - V_m$$

$$\Rightarrow V_m = I_1 Z_{cn} - V_{ab}$$

$$\Rightarrow V_m = 0.324\angle -29^\circ \times (-j318.3) - 200\angle 0^\circ$$

$$\Rightarrow V_m = 265.7\angle -160^\circ \text{ V}$$

Therefore, $V_m > V_L$ (Showed)

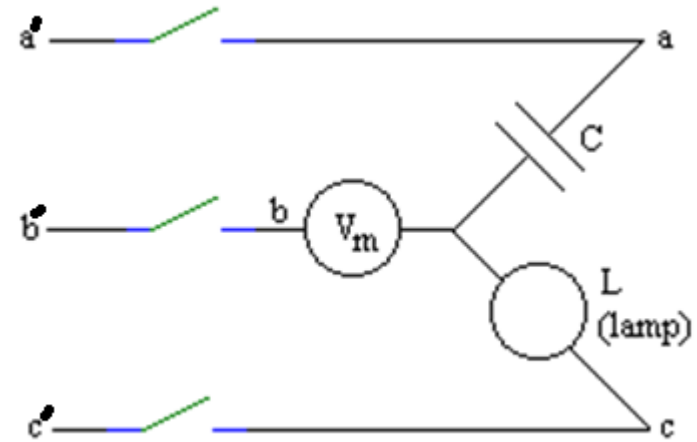
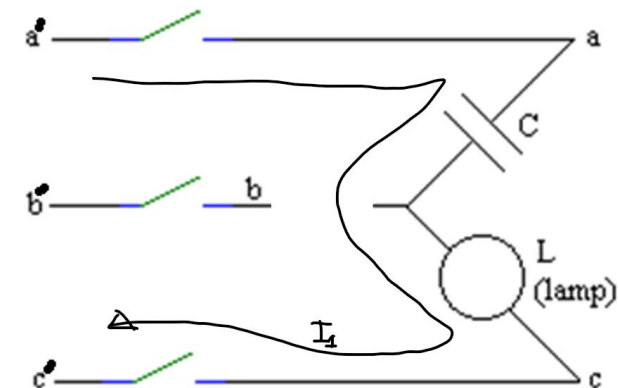


Fig.1



Unbalanced Polyphase Circuits



Q. A 200 V three-phase supply with phase sequence ab–ca–bc is applied to the circuit shown in Fig. 1. The lamp is rated at 230 V, 100 W, and the capacitor has a capacitance of $10\ \mu\text{F}$. Show that the voltmeter reading is less than the line voltage.

Solution:

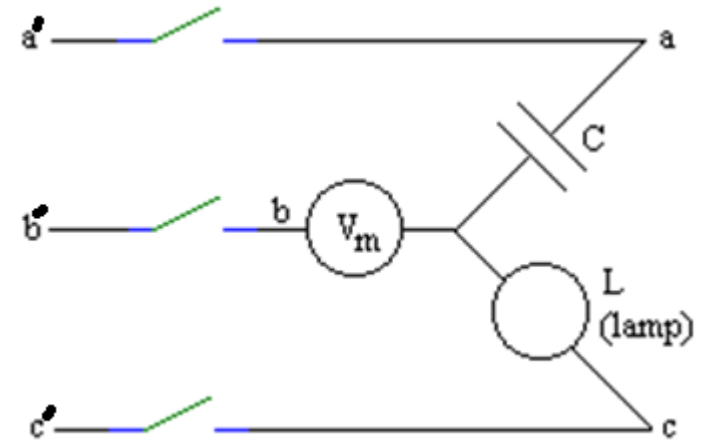


Fig.1

Unbalanced Polyphase Circuits

The Three-Wattmeter Method :

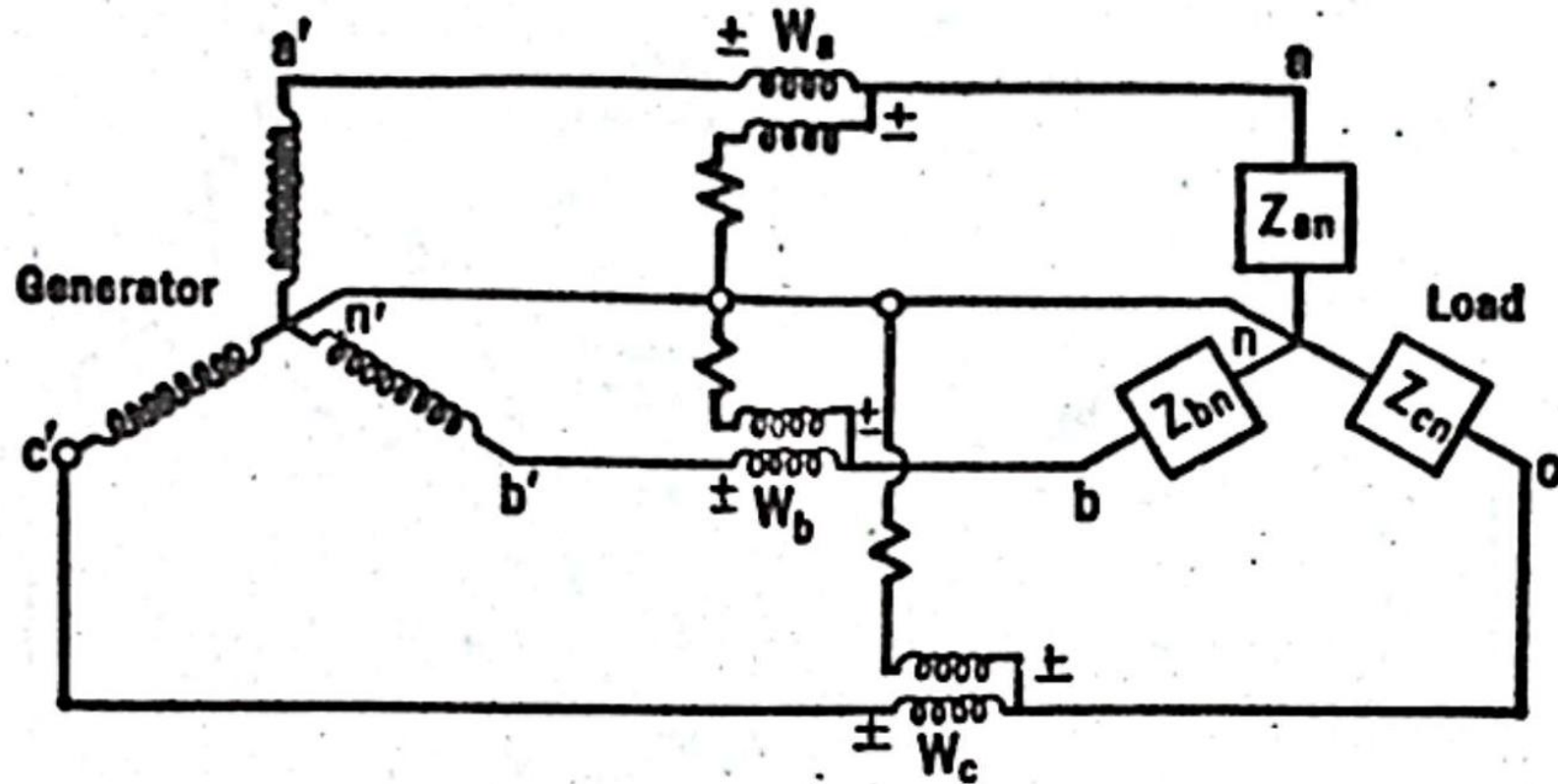


FIG. 12. The three-wattmeter method of measuring four-wire, three-phase power.

Unbalanced Polyphase Circuits

Power Measurement of Unbalanced Loads by the Two-Wattmeter Method:

Q. Refer to Fig. 29. $V_{a'b'}$, $V_{b'c'}$ and $V_{c'a'}$ represent a balanced three-phase system of voltage drops, the magnitude of each being 200 volts. The voltage sequence is $a'b'-b'c'-c'a'$. Two balanced three-phase loads indicated by the circles are connected to the terminals abc as shown in Fig. 29. In addition to the two balanced loads, a single-phase, 4-kw, unity-power-factor load is placed across the bc terminals as indicated. Find the reading of $W_{a'a-ab}$ and $W_{c'c-cb}$.

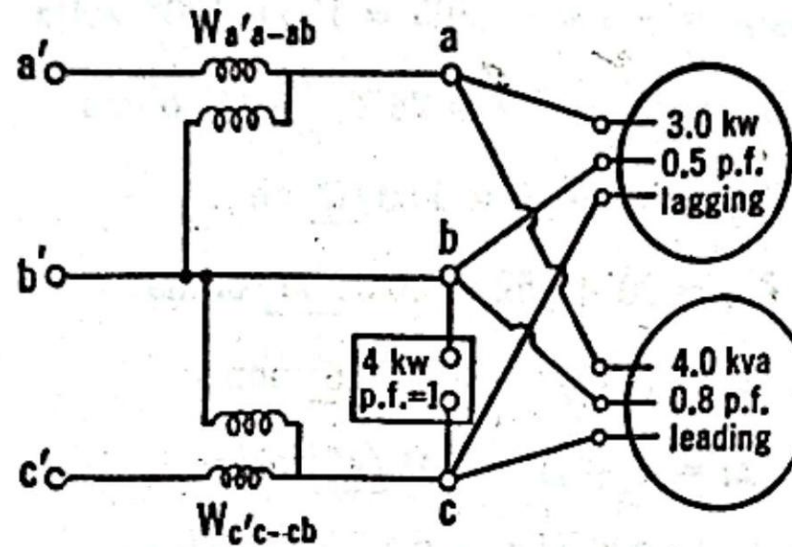


FIG. 29. See Problem 26.

Unbalanced Polyphase Circuits



- From: *Alternating Current Circuits (4th edition)*
Chapter IX(Unbalanced Polyphase Circuits)
- Example:7,10,11,13,
 - problems: 13,28

Enhancing Your Skills and Your Career:

- Developing and enhancing your “ability to identify, formulate, and solve engineering problems” is a primary focus of textbook.
- “An understanding of professional and ethical responsibility” is required of every engineer. To some extent, this understanding is very personal for each of us.
- Your responsibility both professionally and ethically is to perform in a manner that does not harm those around you and to whom you are responsible. Clearly, developing this capability will take time and maturity on your part. I recommend practicing this by looking for professional and ethical components in your day-to-day activities.

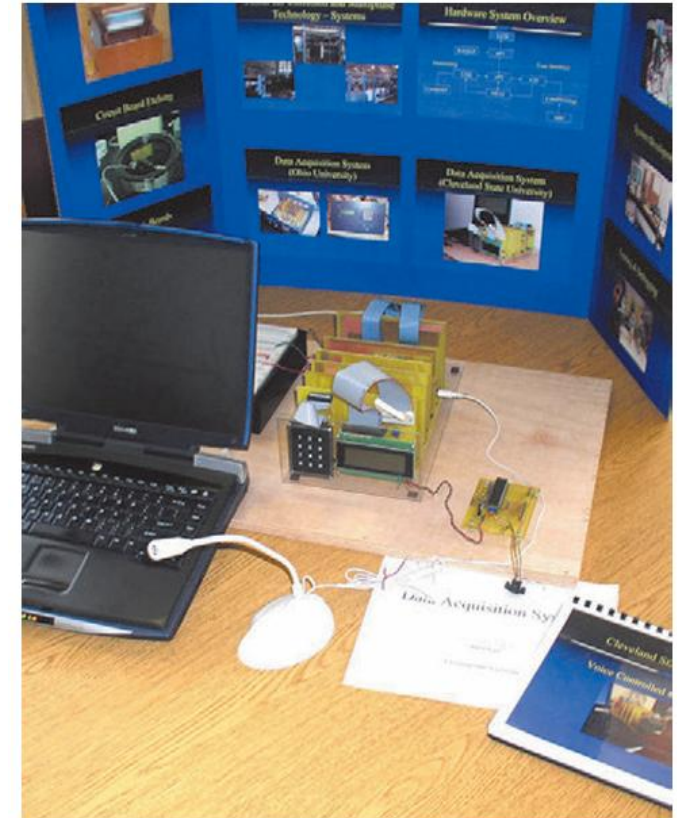


Photo by Charles Alexander

Thank You

